

appears as shown in Fig. 7.

5. Change the X_{AL} (MHz) value to 11.0592 as used in the project. Click

'Output' menu and tick the button against 'Create HEX File' option.

6. Now close the window and go to the main window. Compile the project by clicking 'Build Target' option. The program will be compiled with the message as shown in Fig. 8.

Now, the eftyto.hex code will be generated in the directory where the project file eftyto.uv2 is located. This hex code is used for programming the chip.

Programming the chip using Flash Magic

1. Run the Flash Magic software and select the device as P89V51RD2 and 'Erase all Flash' option as shown in Fig. 9.

2. In the 'Advanced Option,' uncheck 'Use DTR to control RST' and 'Assert DTR and RST while COM port open' options.

3. Browse the eftyto.hex file from your PC and load it onto the main screen. Press 'Start' button to program the chip.

Construction and testing

An actual-size, single-side PCB of the transmitter circuit is shown in Fig. 10 and its

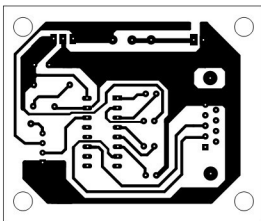


Fig. 10: An actual-size, single-side PCB of the transmitter circuit

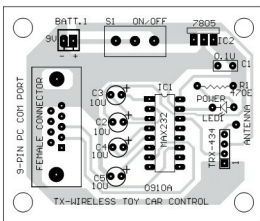


Fig. 11: Component layout for the PCB shown in Fig. 10

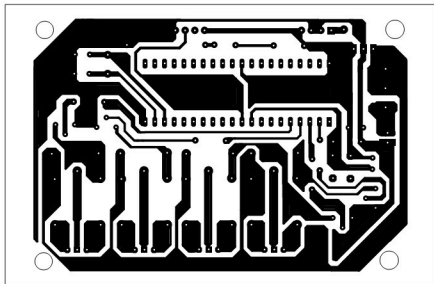


Fig. 12: An actual-size, single-side PCB of the receiver circuit

PARTS LIST

Semiconductors:

IC1	- MAX232 RS-232 level converter
IC2, IC4	- 7805, 5V regulator
IC3	- P89V51RD2 microcontroller
T1-T4	- SL100 npn transistor
TX-434	- 434MHz RF transmitter module
RX-434	- 434MHz RF receiver module
D1-D4	- 1N4007 rectifier diode
LED1, LED2	- 5mm light-emitting diode

Resistors (all 1/4-watt, ±5% carbon):

R1, R2, R4-R7	- 470-ohm
R3	- 8.2-kilo-ohm

Capacitors:

C1, C6	- 0.1µF ceramic
C2-C5, C7	- 10µF electrolytic
C8, C9	- 33pF ceramic

Miscellaneous:

X_{AL}	- 11.0592MHz crystal oscillator
BATT.1	- 9V PP3 battery
BATT.2	- 12V, 4.5Ah, rechargeable SMF battery
S1, S2	- On/off toggle switch
S3	- Tactile switch
RL1-RL4	- 12V, single-changeover relay
M1, M2	- 12V, 50-rpm DC motor
	- Whip antenna
	- 9-pin D-type female connector
	- Metal chassis
	- Nuts and bolts

component layout in Fig. 11. An actual-size, single-side PCB of the receiver circuit is shown in Fig. 12 and its component layout in Fig. 13.

Burn the code into P89V51RD2 and mount the components on the receiver PCB.

Mount the populated PCB, 12V battery and four motors on a suitable chassis, preferably a 20×15cm² metal chassis. Use proper nuts and bolts to fix them firmly on the chassis. Four wheels, each of 7cm dia., are attached to shafts of respective motors.

After mounting the components on the transmitter PCB, connect the circuit to the serial port of the computer using a serial cable. If your PC has a USB port only, you may use a USB-to-serial converter. Now switch on the 9V power supply using switch S1.

Before sending the data from your computer, check which com port is connected to the circuit. Set the baud rate of the com port to 1200, stop bit to 1 and data bits to 8. Go to 'Device Manager' option from your desktop to do these settings. Now run a serial communica-